

Unit 9

Correlation, Regression, and Calibration Curves

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Scatter plots, correlation, linear regression, residuals, calibration curves, prediction, and uncertainty.
Central habit	Do not report only R^2 . Use the graph, equation, residuals, uncertainty, and scientific context together.

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1 Purpose of Unit 9

Many IB investigations do not compare groups. Instead, they ask whether one measured quantity changes with another:

- absorbance vs concentration in a calibration curve;
- temperature vs enzyme activity;
- light intensity vs photosynthesis rate;
- substrate concentration vs reaction rate;
- time vs product formation;
- pH vs growth, activity, or reaction yield.

These questions require students to think about **relationships**, not only differences between group means.

Core thesis

A regression line is not just a line drawn through points. It is a model that turns data into an equation, a prediction, and a scientific claim. That model must be checked before it is trusted.

By the end of this unit, students should be able to:

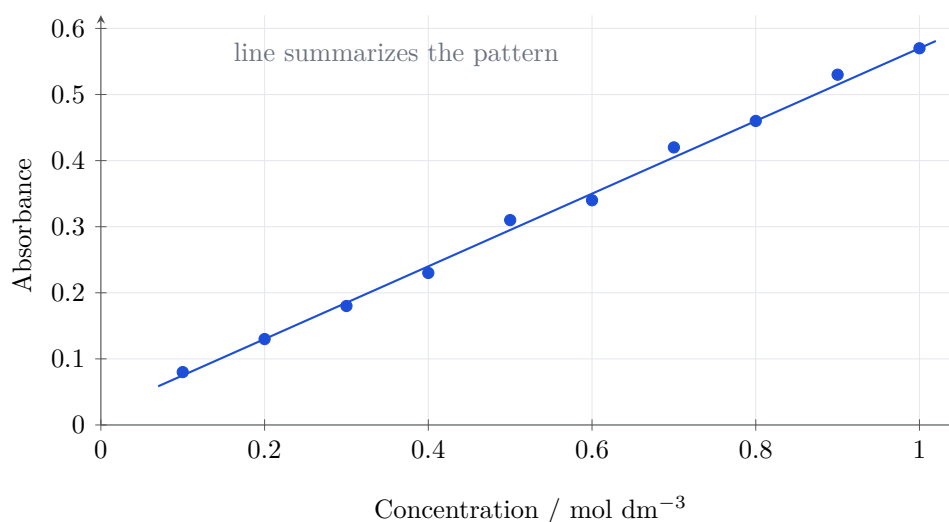
1. choose an appropriate graph for two quantitative variables;
2. distinguish correlation from regression;
3. explain Pearson's r , Spearman's ρ , slope, intercept, and R^2 ;
4. calculate and interpret residuals;
5. use a linear calibration curve to estimate an unknown concentration;
6. explain why extrapolation is dangerous;
7. identify when a relationship is clearly nonlinear;
8. report regression evidence without overclaiming.

2 Start with the scatter plot

A scatter plot is the first graph for two quantitative variables. It shows whether a relationship is increasing, decreasing, curved, noisy, clustered, or driven by one unusual point.

Chemistry example

A student prepares copper sulfate standards and measures absorbance using a colorimeter. The research question is: how does concentration affect absorbance within the tested range?



Quick check

Before calculating anything, answer from the plot:

1. Is the relationship positive, negative, or unclear?
2. Does a straight line seem reasonable over this range?
3. Is there any point that seems unusually far from the pattern?

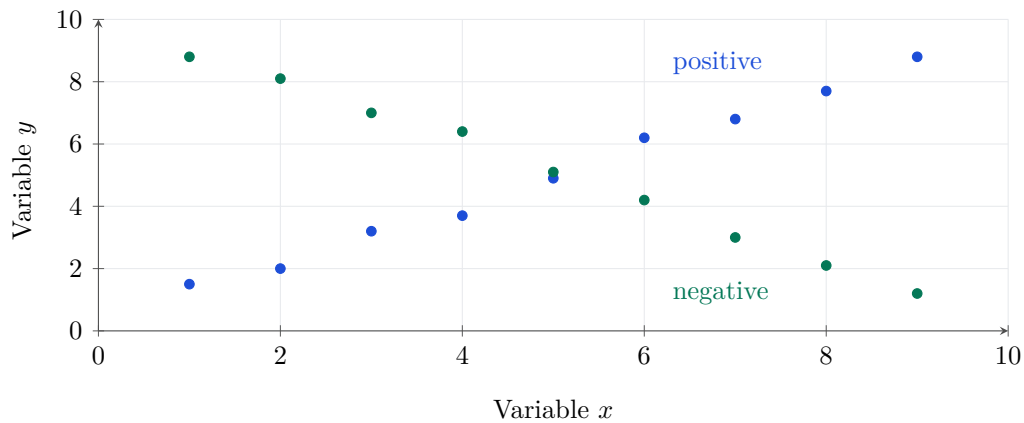
3 Correlation is not the same as regression

Correlation measures the strength and direction of association between two variables. **Regression** fits an equation for predicting or explaining one variable from another.

Question	Correlation	Regression
Main purpose	Measure association	Fit an equation
Variables	Usually symmetric: x and y are treated similarly	Directional: response y is modelled using predictor x
Typical output	r or ρ	slope, intercept, residuals, R^2
IB use	Describe strength of relationship	Calibration curves, prediction, trends, rates

Causation warning

A strong correlation does not prove that one variable causes the other. Causation depends on design, controls, mechanism, and alternative explanations.



4 Pearson's r and Spearman's ρ

Pearson's correlation coefficient r measures the strength of a **linear** relationship.

$$-1 \leq r \leq 1$$

Value of r	Interpretation
$r \approx +1$	Strong positive linear relationship
$r \approx 0$	No strong linear relationship
$r \approx -1$	Strong negative linear relationship

Spearman's rank correlation ρ measures whether values tend to increase or decrease together in a monotonic way. It is based on ranks rather than raw values.

When Spearman is useful

Spearman's ρ is useful when the relationship is increasing or decreasing but not clearly linear, or when ordinal ranks are more appropriate than raw numerical distances.

Do not overinterpret correlation size

A high correlation can occur in a narrow artificial range, from shared confounding variables, or from a small number of points. Always inspect the scatter plot.

5 The linear regression model

A simple linear regression model has this form:

Linear model

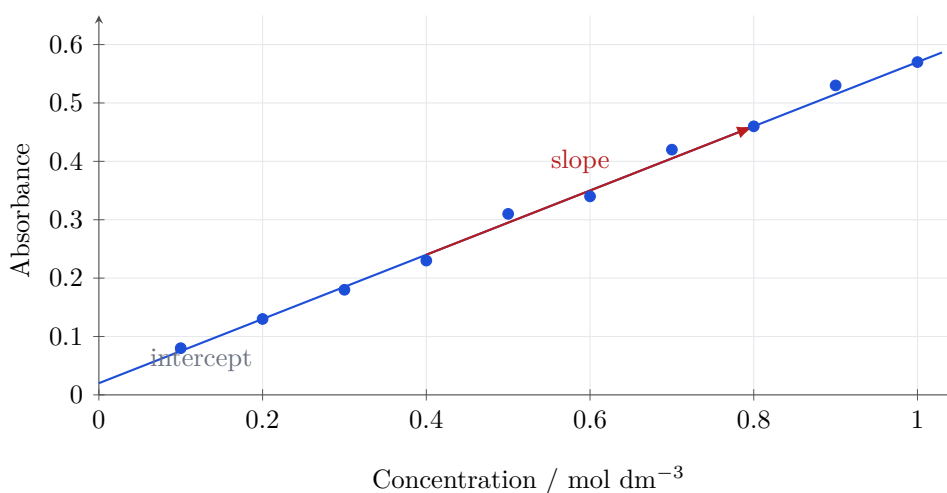
$$\hat{y} = a + bx$$

- $\hat{y}$ is the predicted value of the response variable.
- a is the intercept: the predicted value of y when $x = 0$.
- b is the slope: the predicted change in y for a one-unit increase in x .

For a calibration curve:

$$\text{Absorbance} = a + b(\text{concentration})$$

The slope is often the most important number because it describes how sensitively the measured signal changes with concentration.



6 Residuals: the model errors

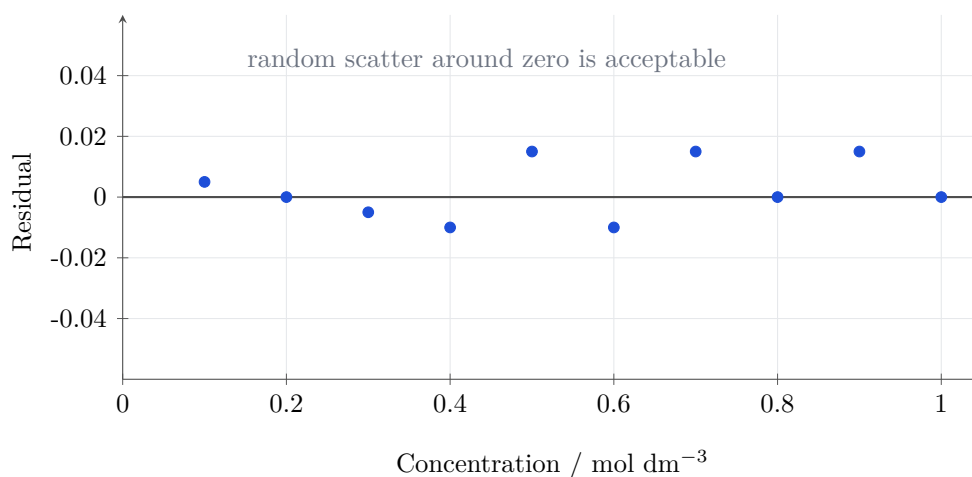
A residual is the difference between an observed value and the value predicted by the model.

Residual

$$e_i = y_i - \hat{y}_i$$

A positive residual means the observed point is above the fitted line. A negative residual means it is below the fitted line.

Residuals are not decorative. They tell you whether the model is leaving a pattern behind.



Residual warning

If residuals form a curve, fan shape, or cluster pattern, the straight-line model may be inappropriate even if R^2 looks high.

7 What R^2 does and does not mean

R^2 is the coefficient of determination. In simple terms, it describes the proportion of variation in the response variable that is explained by the fitted model.

Coefficient of determination

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

where SSE is the residual sum of squares and SST is the total sum of squares.

A high R^2 can be useful, but it does not prove that the model is scientifically correct.

Weak interpretation	Better interpretation
$R^2 = 0.98$, so the experiment was accurate.	The fitted model explains 98% of the variation in this dataset. Accuracy requires calibration, controls, and uncertainty analysis.
High R^2 proves linearity.	A high R^2 can still occur with curved data over a narrow range. Check residuals.
Low R^2 means no relationship.	The relationship may be nonlinear, noisy, range-limited, or affected by uncontrolled variation.

Quick check

A student reports only: "The graph had $R^2 = 0.991$, therefore the unknown concentration is reliable." What extra information should be requested before trusting the claim?

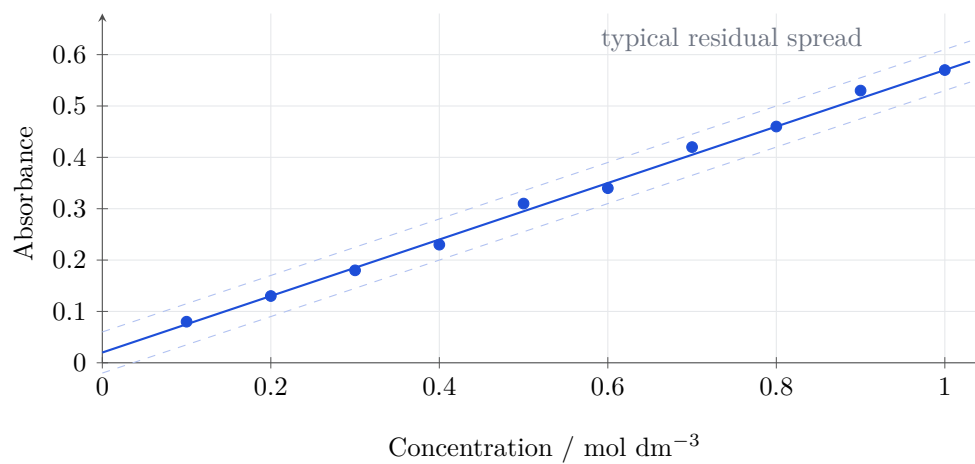
8 Standard error of regression

The standard error of regression measures the typical vertical size of the residuals around the fitted line.

Residual standard error

$$s_{y|x} = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

For a calibration curve, this tells us the typical error in absorbance units around the line. Smaller residual error means predictions from the line are usually more precise, provided the model is appropriate.



9 Calibration curves and unknown concentration

A calibration curve uses standards of known concentration to estimate an unknown sample.

Calibration logic

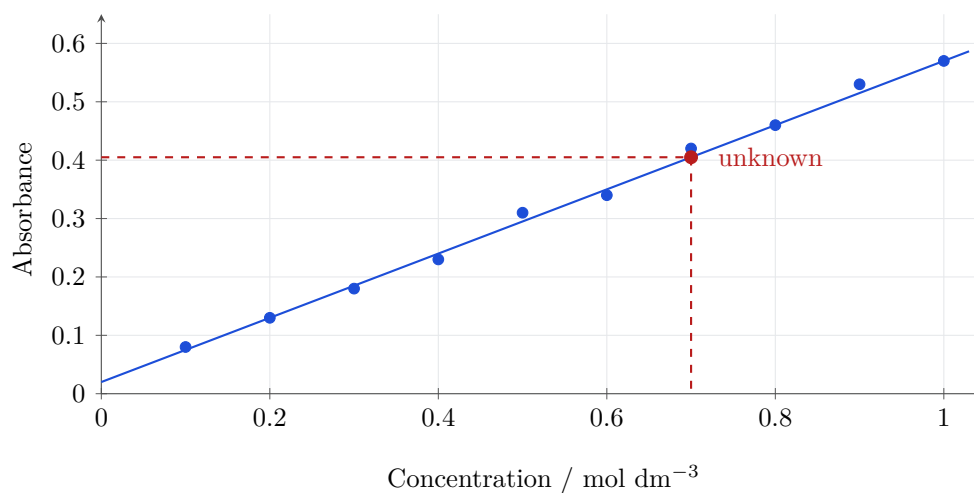
Standards establish the relationship between concentration and signal. The unknown is then estimated by placing its measured signal onto the fitted model.

Suppose the fitted calibration equation is:

$$A = 0.020 + 0.550C$$

where A is absorbance and C is concentration in mol dm^{-3} . If an unknown sample has absorbance $A = 0.405$, then:

$$C_{\text{unknown}} = \frac{A - a}{b} = \frac{0.405 - 0.020}{0.550} = 0.700 \text{ mol dm}^{-3}$$

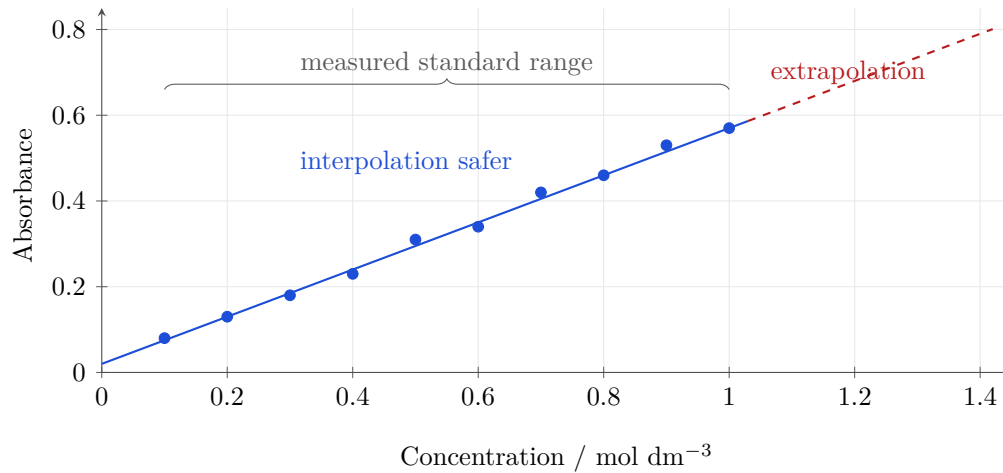


Calibration warning

The unknown should fall inside the range of standards. If it is above the highest standard or below the lowest standard, dilute the sample or prepare a better standard range.

10 Interpolation vs extrapolation

Interpolation means estimating within the measured range. **Extrapolation** means estimating outside the measured range. Interpolation is usually much safer.



Why extrapolation is dangerous

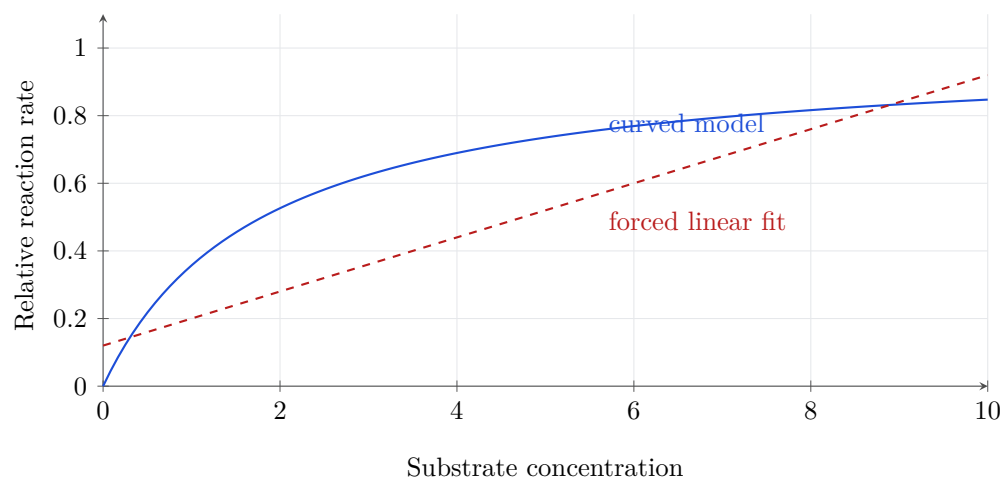
The relationship may stop being linear outside the tested range. Instruments can saturate, enzymes can reach a maximum rate, reagents can become limiting, and biological systems can shift mechanism.

11 Nonlinear relationships

Not every relationship should be forced into a straight line. In biology especially, many relationships curve.

Biology example

Enzyme activity often increases with substrate concentration at low concentration, then approaches a maximum. A straight line may be reasonable only over a small initial region.



Practical rule

If the scatter plot visibly curves, do not hide the curvature with a linear R^2 . Either restrict the analysis to a justified linear region or use a model appropriate for the process.

12 Correlation, regression, and causation

Regression can support a causal interpretation only when the experimental design supports it. A strong line alone is not enough.

Situation	What can be claimed?
Controlled experiment where only temperature was changed	Temperature is a plausible cause of the observed change, if controls were maintained.
Observational data on student height and lung capacity	There is an association, but causation requires caution because age, sex, training, and body size may confound the pattern.
Calibration standards prepared from known concentrations	Regression can be used for estimation within the validated range.
One outlier creates most of the slope	The relationship is fragile; inspect and justify treatment of the point.

Bad wording

Avoid: "The high R^2 proves that concentration caused absorbance." Better: "Within the tested standard range, absorbance increased approximately linearly with concentration, supporting use of a calibration curve."

13 Reporting regression in IB science

A useful regression conclusion should mention the pattern, equation, effect size, uncertainty, model checks, and limitations.

Good reporting template

The scatter plot showed an approximately [linear/curved] relationship between [predictor] and [response]. A linear model gave $\hat{y} = a + bx$, with slope $b = \dots$. The residual plot [did/did not] show systematic curvature. The model was used only within the tested range of standards. The main limitation was [measurement uncertainty / small sample size / narrow range / possible confounding].

Report this	Because it answers
Scatter plot	What pattern is visible?
Slope and units	How large is the effect per unit change in x ?
Intercept	Is it meaningful or just a mathematical fitting parameter?
R^2	How much variation does the model explain?
Residual plot	Is a straight line reasonable?
Calibration range	Is the unknown an interpolation or extrapolation?
Uncertainty	How precise is the estimate or prediction?

14 Worked mini-example

A student measures absorbance for standards and obtains the following fitted equation:

$$A = 0.018 + 0.612C$$

The unknown sample gives:

$$A = 0.385$$

Estimated concentration:

$$C = \frac{0.385 - 0.018}{0.612} = 0.600 \text{ mol dm}^{-3}$$

Interpretation task

Write a cautious conclusion using the following information:

- the unknown falls within the standard range;
- $R^2 = 0.987$;
- residuals show no obvious curve;
- pipetting uncertainty and colorimeter drift may affect precision.

15 Common mistakes

Mistake	Better habit
Reporting only R^2 .	Report slope, equation, residuals, range, and uncertainty.
Using a bar chart for two quantitative variables.	Use a scatter plot.
Extrapolating far beyond the standards.	Prepare new standards or dilute the unknown.
Calling correlation causation.	Discuss design and possible confounders.
Ignoring residuals.	Use residuals to check whether a linear model is reasonable.
Treating the intercept as meaningful automatically.	Ask whether $x = 0$ is physically meaningful and within the studied range.

16 Unit 9 summary

Essential takeaways

- Scatter plots come before correlation or regression.
- Correlation measures association; regression fits a predictive model.
- The slope is often more scientifically meaningful than R^2 .
- Residuals reveal whether the model is leaving structure unexplained.
- Calibration curves should be used inside the standard range whenever possible.
- A high R^2 does not prove causation, accuracy, or model validity.